

Assignment 3: **Text Classification with LLM**

In the previous assignment, you did text classification using fully connected neural network, LSTM and also used pre-trained word embedding. You went through the process of vectorisation and many other pre-processing steps. This was state-of-the-art, up until LLM were introduced. Now most of the NLP related task are implemented using LLMs. During class we practiced couple of sentiment analysis tasks using LLM. In this assignment, your tasks are,

Using transformer's Encoder only, train a model on 80% of the data from bbcports dataset and report accuracy on remaining on 20% data

Draft an appropriate LLM prompt, which includes correct system, user and assistant roles and context

Choose at least five LLMs (deepseek, chatgpt, lama, bloom, etc..) Try playing with different LLMs, see which of the LLMs are open source, which LLMs are from Meta, OpenAI, Google etc. You should play with a handful of LLMs and finalise five LLMs to perform classification on bbcports dataset

It is obvious that you won't be able to perform classification on full dataset, but at least, in order to appreciate the power LLMs, select 200 instances, evenly distributed in class labels and then perform classification with all five LLM

You should do everything in one notebook, you might need to do it locally on your machine if you are unable to configure deepseek on colab

Write a proper conclusion, which LLM performs best.

Also write at least three lesson learnt from this assignment.

Once you are done, locally or on colab, download ipynb file and upload on the submission of this assignment.